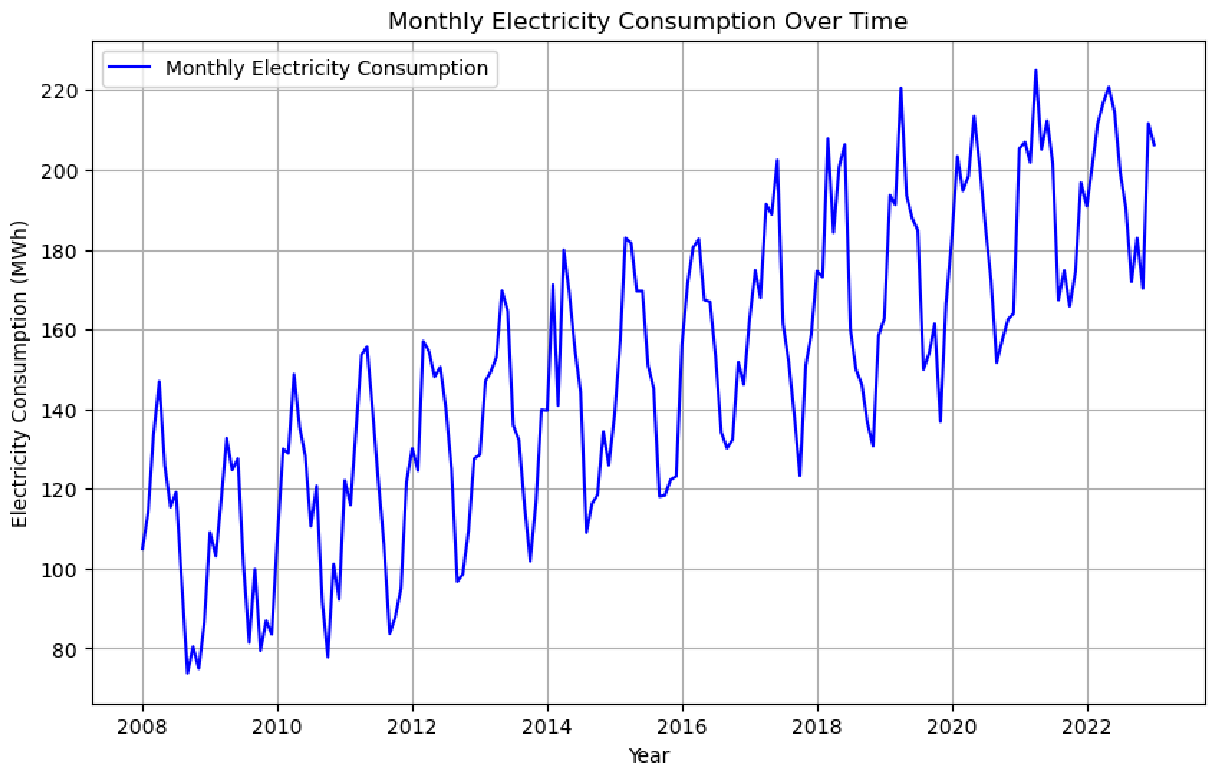
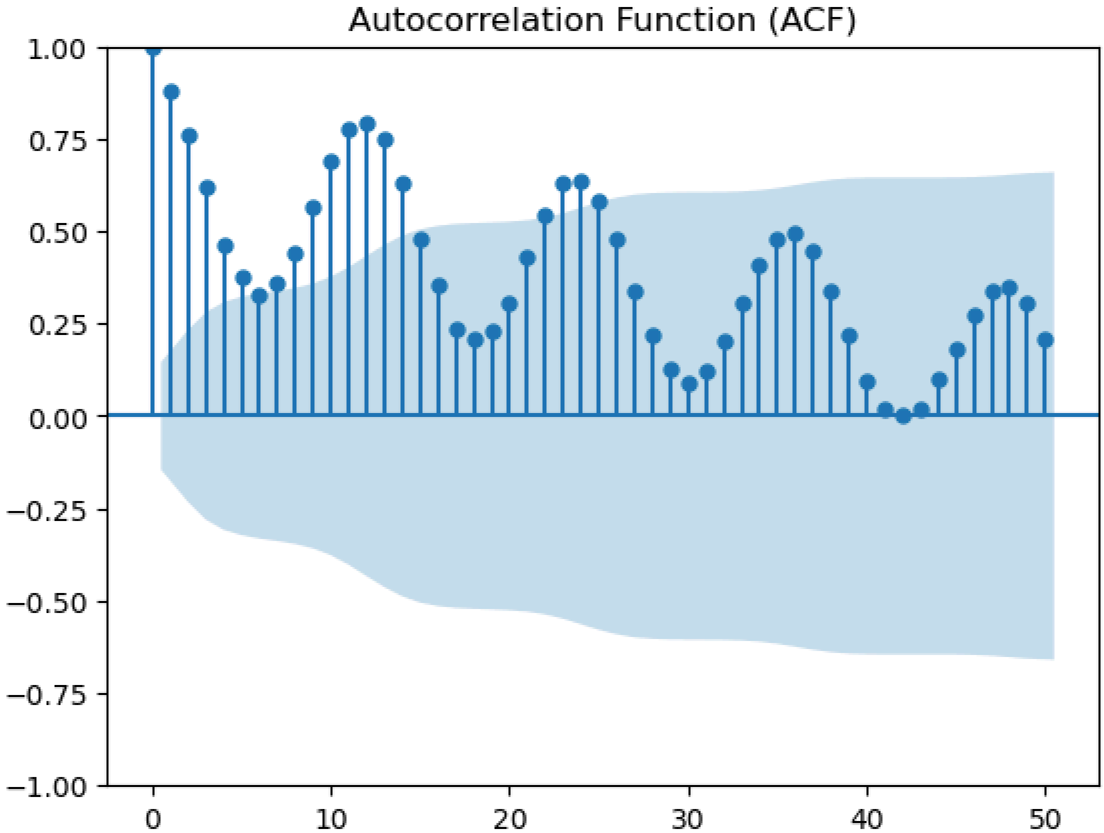
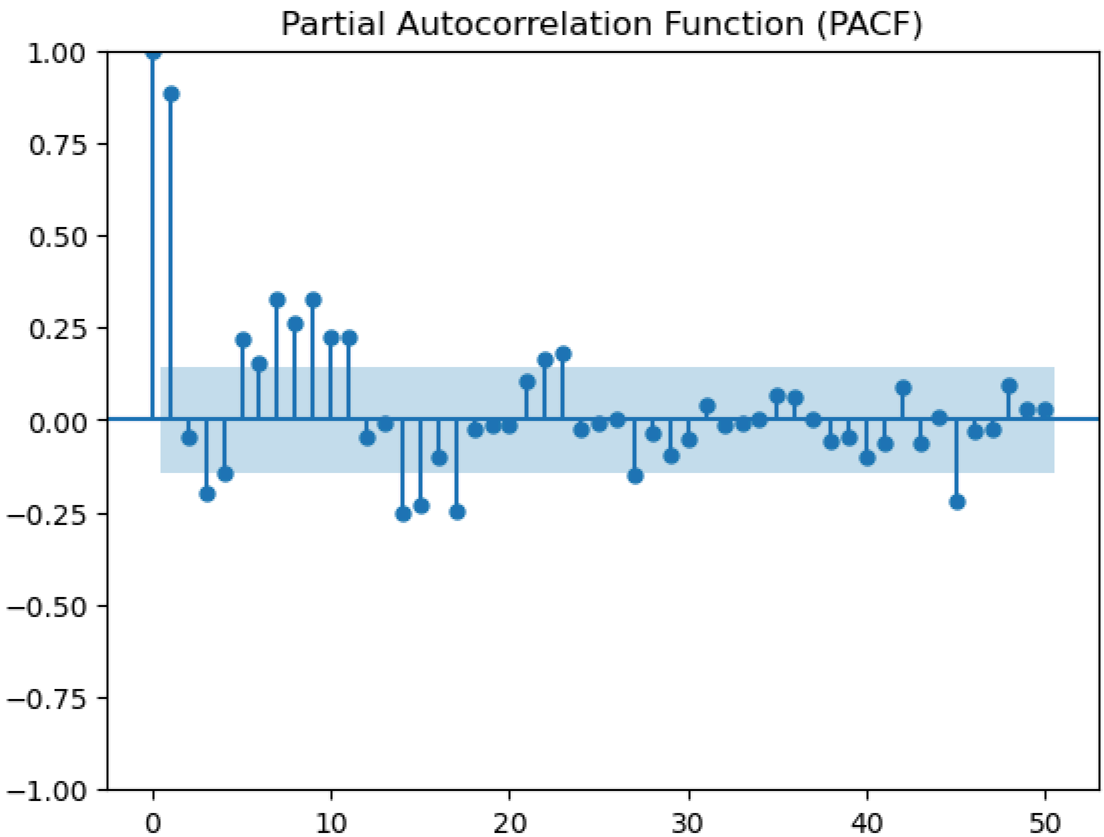
	Date	Consumption
0	2008-01-01	104.97
1	2008-02-01	114.17
2	2008-03-01	133.57
3	2008-04-01	146.90
4	2008-05-01	125.86






```
C:\Users\admin\anaconda3\Lib\site-packages\statsmodels\graphics\tsaplots.py:
348: FutureWarning: The default method 'yw' can produce PACF values outside
of the [-1,1] interval. After 0.13, the default will change tounadjusted Yul
e-Walker ('ywm'). You can use this method now by setting method='ywm'.
  warnings.warn(
```



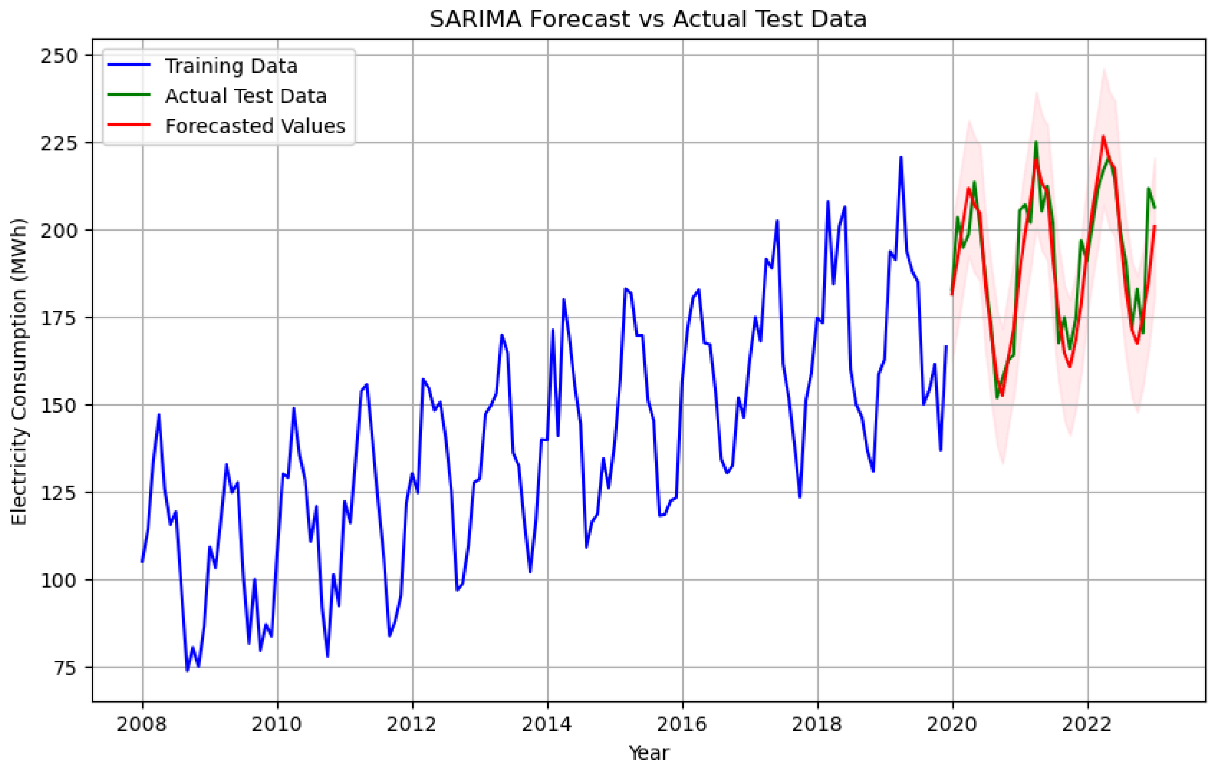

```
C:\Users\admin\anaconda3\Lib\site-packages\statsmodels\tsa\base\tsa_model.p
y:471: ValueWarning: No frequency information was provided, so inferred freq
uency MS will be used.
  self._init_dates(dates, freq)
C:\Users\admin\anaconda3\Lib\site-packages\statsmodels\tsa\base\tsa_model.p
y:471: ValueWarning: No frequency information was provided, so inferred freq
uency MS will be used.
  self._init_dates(dates, freq)
C:\Users\admin\anaconda3\Lib\site-packages\statsmodels\tsa\statespace\sarima
x.py:966: UserWarning: Non-stationary starting autoregressive parameters fou
nd. Using zeros as starting parameters.
  warn('Non-stationary starting autoregressive parameters'
C:\Users\admin\anaconda3\Lib\site-packages\statsmodels\tsa\statespace\sarima
x.py:978: UserWarning: Non-invertible starting MA parameters found. Using ze
ros as starting parameters.
  warn('Non-invertible starting MA parameters found.'
```

```
                               SARIMAX Results                                
==============================================================================
==============
Dep. Variable:            Electricity_Consumption   No. Observations:         
144
Model:             SARIMAX(1, 0, 1)x(1, 1, 1, 12)   Log Likelihood            
-501.884
Date:                            Tue, 01 Apr 2025   AIC                       
1013.768
Time:                                    11:38:43   BIC                       
1028.182
Sample:                                01-01-2008   HQIC                      
1019.625
                                     - 12-01-2019                                         
Covariance Type:                              opg                                         
==============================================================================
==
                 coef    std err          z      P>|z|      [0.025      0.97
5]
------------------------------------------------------------------------------
--
ar.L1          1.0000   4.99e-06      2e+05      0.000       1.000       1.0
00
ma.L1         -0.9867      0.149     -6.620      0.000      -1.279      -0.6
95
ar.S.L12      -0.0870      0.091     -0.960      0.337      -0.265       0.0
91
ma.S.L12      -0.9589      0.178     -5.384      0.000      -1.308      -0.6
10
sigma2        91.9150      0.002   5.32e+04      0.000      91.912      91.9
18
==============================================================================
=======
Ljung-Box (L1) (Q):                   1.58   Jarque-Bera (JB):                
0.03
Prob(Q):                              0.21   Prob(JB):                        
0.99
Heteroskedasticity (H):               1.25   Skew:                            
0.02
Prob(H) (two-sided):                  0.47   Kurtosis:                        
2.95
==============================================================================
=======

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (compl
ex-step).
[2] Covariance matrix is singular or near-singular, with condition number 1.
64e+20. Standard errors may be unstable.
```

```
C:\Users\admin\anaconda3\Lib\site-packages\statsmodels\base\model.py:604: Co
nvergenceWarning: Maximum Likelihood optimization failed to converge. Check 
mle_retvals
  warnings.warn("Maximum Likelihood optimization failed to "
```

```
Root Mean Squared Error (RMSE): 8.99
```


SARIMA Forecast vs Actual Test Data
Training Data
Actual Test Data
Forecasted Values
Electricity Consumption (MWh)
Year
250
225
200
175
150
125
100
75
2008
2010
2012
2014
2016
2018
2020
2022